



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目: 高等数学(下) 开课单位: 数学系

考试时长: 120 分钟 命题教师: _____

题号	1	2	3	4	5	6	7	8
分值	20 分	20 分	10 分	10 分	10 分	10 分	10 分	10 分

本试卷共 8 道大题, 满分 100 分. (考试结束后请将试卷、答题本、草稿纸一起交给监考老师)

注意: 本试卷里的中文为直译 (即完全按英文字面意思直接翻译), 所有数学词汇的定义请参照教材 (Thomas' Calculus, 13th Edition) 中的定义。如果其中有些数学词汇的定义不同于中文书籍 (比方说同济大学的高等数学教材) 里的定义, 以教材 (Thomas' Calculus, 13th Edition) 中的定义为准。

1. (20pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

(1) Which of the following vector is orthogonal to the tangent line for the curve

$$\mathbf{r}(t) = (1 + 1/t^2)\mathbf{i} + (1 - 3/t)\mathbf{j} + t^2\mathbf{k}$$

at the point $P = (2, -2, 1)$?

(A) $\mathbf{v} = \langle 1, 2, 1 \rangle$.

(B) $\mathbf{v} = \langle 1, 2, 2 \rangle$.

(C) $\mathbf{v} = \langle -1, -2, 2 \rangle$.

(D) $\mathbf{v} = \langle -1, -2, -2 \rangle$.

(2) The derivative of $f(x, y, z) = x^2y + z^2$ at the point $(1, 2, 0)$ in the direction of $\mathbf{v} = \langle 1, 2, 2 \rangle$ is

(A) 12.

(B) 6.

(C) 4.

(D) 2.

(3) The series

$$\sum_{n=1}^{\infty} \left(1 - \frac{a}{n}\right)^{n^2}$$

converges

(A) for any real number a .

(B) for any $a \in (-1, 1)$.

(C) for any $a > 0$.

(D) for any $a < 0$.

- (4) $\lim_{(x,y) \rightarrow (0,0)} (1 + \sin(xy))^{\frac{1}{x^2+y^2}} =$
 (A) 0 . (B) 1 .
 (C) e . (D) does not exist.

- (5) $\int_0^{\pi/4} \int_0^{2\cos\theta} f(r\cos\theta, r\sin\theta) r dr d\theta =$
 (A) $\int_0^1 \int_y^{\sqrt{2y-y^2}} f(x, y) dx dy.$
 (B) $\int_0^1 \int_{1-\sqrt{1-y^2}}^y f(x, y) dx dy.$
 (C) $\int_0^1 \int_{-x}^x f(x, y) dy dx + \int_1^2 \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} f(x, y) dy dx.$
 (D) $\int_0^1 \int_0^x f(x, y) dy dx + \int_1^2 \int_0^{\sqrt{2x-x^2}} f(x, y) dy dx.$

2. (20 pts) Fill in the blanks.

- (1) The equation of plane tangent to the surface $xyz + e^{x+z} = 0$ at the point $(1, 1, -1)$ is _____.
- (2) If the Maclaurin series of a function $f(x)$ is $\sum_{n=1}^{\infty} \frac{x^{2n+2}}{n(n+4)2^n}$, then $f^{(6)}(0) =$ _____.
- (3) The curvature κ for the curve $\mathbf{r}(t) = e^t \cos t \mathbf{i} + e^t \sin t \mathbf{j} + 2\mathbf{k}$ is _____.
- (4) If a function $y = y(x)$ is defined by the parametric equations

$$x = 2t + \ln^2 t, \quad y = (t + \ln t)^2, \quad t > 0,$$

then $\left. \frac{d^2 y}{dx^2} \right|_{t=1} =$ _____.

(5) $\int_0^1 \int_x^1 x e^{-y^3} dy dx =$ _____.

3. (10 pts) Find all the local extreme values of $f(x, y) = x^3 + 2y^2 - 3x - 12y$.

4. (10 pts)

- (1) Find the interval of convergence of the series $\sum_{n=2}^{\infty} \frac{(-1)^n (x-1)^{2n+1}}{(n+2023) \ln n}$.
- (2) For what values of x does the series converge absolutely, or conditionally?

5. (10 pts) Find the center of mass of the hemisphere $D: x^2 + y^2 + z^2 \leq 1, z \geq 0$ with the density function $\delta(x, y, z) = z$.

6. (10 pts) Given the conservative vector field

$$\mathbf{F} = (y \sin z + 2)\mathbf{i} + (x \sin z)\mathbf{j} + (xy \cos z)\mathbf{k},$$

find its potential function and evaluate the integral

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

along any smooth curve C joining point $A(1, -2, \pi/3)$ to $B(2, 1, \pi/4)$.

7. (10 pts) The vector-field

$$\mathbf{F}(x, y, z) = (z + xy^2)\mathbf{i} + (2y + y^3)\mathbf{j} - 4zy^2\mathbf{k}$$

is defined in the domain V bounded by the surfaces S_1 and S_2 ,

$$S_1 := \{(x, y, z) : z = (x^2 + y^2)^2 - 1\},$$

$$S_2 := \{(x, y, z) : z = 4 - 4(x^2 + y^2)\}.$$

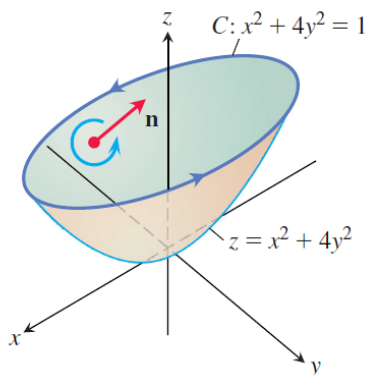
Apply the **Divergence Theorem** to compute the outward flux of $\mathbf{F}$ across the boundary of V .

8. (10 pts) Let the surface S be the elliptical paraboloid $z = x^2 + 4y^2$ lying beneath the plane $z = 1$ (Figure is given below). We define the orientation of S by taking the inner normal vector $\mathbf{n}$ to the surface, which is the normal having a positive $\mathbf{k}$ -component. Compute

$$\iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} \, d\sigma,$$

here

$$\mathbf{F} = (y + x^2 \ln(x^4 + 1))\mathbf{i} + (e^{\sin y} - xz)\mathbf{j} + (xz^2 + \cos(z^2 + 1))\mathbf{k}.$$



一、 (20分) 单项选择题:

(1) 下列哪一个向量与曲线

$$\mathbf{r}(t) = (1 + 1/t^2)\mathbf{i} + (1 - 3/t)\mathbf{j} + t^2\mathbf{k}$$

在点 $P = (2, -2, 1)$ 处的切线垂直?

- (A) $\mathbf{v} = \langle 1, 2, 1 \rangle$. (B) $\mathbf{v} = \langle 1, 2, 2 \rangle$.
(C) $\mathbf{v} = \langle -1, -2, 2 \rangle$. (D) $\mathbf{v} = \langle -1, -2, -2 \rangle$.

(2) 函数 $f(x, y, z) = x^2y + z^2$ 在点 $(1, 2, 0)$ 处沿向量 $\mathbf{v} = \langle 1, 2, 2 \rangle$ 的方向的方向导数是

- (A) 12. (B) 6.
(C) 4. (D) 2.

(3) 级数

$$\sum_{n=1}^{\infty} \left(1 - \frac{a}{n}\right)^{n^2}$$

- (A) 对任意实数 a 都收敛.
(B) 当 $a \in (-1, 1)$ 时收敛.
(C) 当 a 为任意正实数时收敛.
(D) 当 a 为任意负实数时收敛.

(4) $\lim_{(x,y) \rightarrow (0,0)} (1 + \sin(xy))^{\frac{1}{x^2+y^2}} =$

- (A) 0. (B) 1.
(C) e . (D) 极限不存在.

(5) $\int_0^{\pi/4} \int_0^{2\cos\theta} f(r\cos\theta, r\sin\theta) r dr d\theta =$

- (A) $\int_0^1 \int_y^{\sqrt{2y-y^2}} f(x, y) dx dy$.
(B) $\int_0^1 \int_{1-\sqrt{1-y^2}}^y f(x, y) dx dy$.
(C) $\int_0^1 \int_{-x}^x f(x, y) dy dx + \int_1^2 \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} f(x, y) dy dx$.
(D) $\int_0^1 \int_0^x f(x, y) dy dx + \int_1^2 \int_0^{\sqrt{2x-x^2}} f(x, y) dy dx$.

二、 (20分) 填空题:

(1) 曲面 $xyz + e^{x+z} = 0$ 在点 $(1, 1, -1)$ 处的切平面的方程是_____.

(2) 若函数 $f(x)$ 的麦克劳林级数为 $\sum_{n=1}^{\infty} \frac{x^{2n+2}}{n(n+4)2^n}$, 则 $f^{(6)}(0) =$ _____.

(3) 曲线 $\mathbf{r}(t) = e^t \cos t \mathbf{i} + e^t \sin t \mathbf{j} + 2\mathbf{k}$ 的曲率 $\kappa =$ _____.

(4) 若函数 $y = y(x)$ 的参数方程为

$$x = 2t + \ln^2 t, \quad y = (t + \ln t)^2, \quad t > 0,$$

则 $\left. \frac{d^2 y}{dx^2} \right|_{t=1} = \underline{\hspace{2cm}}$.

(5) $\int_0^1 \int_x^1 x e^{-y^3} dy dx = \underline{\hspace{2cm}}$.

三、(10分) 求函数 $f(x, y) = x^3 + 2y^2 - 3x - 12y$ 的所有极值.

四、(10分)

(1) 求级数 $\sum_{n=2}^{\infty} \frac{(-1)^n (x-1)^{2n+1}}{(n+2023) \ln n}$ 的收敛域.

(2) x 取哪些值时上述级数绝对收敛, 取哪些值时条件收敛.

五、(10分) 若上半球 $D: x^2 + y^2 + z^2 \leq 1, z \geq 0$ 的密度函数为 $\delta(x, y, z) = z$, 计算此上半球的质心.

六、(10分) 已知向量场

$$\mathbf{F} = (y \sin z + 2)\mathbf{i} + (x \sin z)\mathbf{j} + (xy \cos z)\mathbf{k},$$

是保守场, 求 $\mathbf{F}$ 的势函数, 并计算曲线积分

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

这里 C 是从点 $A(1, -2, \pi/3)$ 到点 $B(2, 1, \pi/4)$ 的光滑曲线.

七、(10分) 向量场

$$\mathbf{F}(x, y, z) = (z + xy^2)\mathbf{i} + (2y + y^3)\mathbf{j} - 4zy^2\mathbf{k}$$

的定义域 V 是夹在如下曲面 S_1 和 S_2 之间的闭区域, 这里

$$S_1 := \{(x, y, z) : z = (x^2 + y^2)^2 - 1\},$$

$$S_2 := \{(x, y, z) : z = 4 - 4(x^2 + y^2)\}.$$

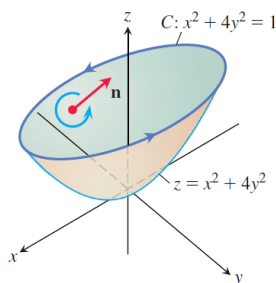
使用**散度定理**来计算向量场 $\mathbf{F}$ 通过 V 的边界从内向外的通量(flux).

八、(10分) 设曲面 S 是椭圆抛物面 $z = x^2 + 4y^2$ 在平面 $z = 1$ 下方的部分. 曲面 S 的内法向量 $\mathbf{n}$ 的方向如下图所示. 计算

$$\iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} d\sigma,$$

这里

$$\mathbf{F} = (y + x^2 \ln(x^4 + 1))\mathbf{i} + (e^{\sin y} - xz)\mathbf{j} + (xz^2 + \cos(z^2 + 1))\mathbf{k}.$$



1. (20pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

C (1) Which of the following vector is orthogonal to the tangent line for the curve

$$\mathbf{r}(t) = (1 + 1/t^2)\mathbf{i} + (1 - 3/t)\mathbf{j} + t^2\mathbf{k}$$

at the point $P = (2, -2, 1)$?

(A) $\mathbf{v} = \langle 1, 2, 1 \rangle$.

(B) $\mathbf{v} = \langle 1, 2, 2 \rangle$.

(C) $\mathbf{v} = \langle -1, -2, 2 \rangle$.

(D) $\mathbf{v} = \langle -1, -2, -2 \rangle$.

$$\mathbf{r}'(t) = -\frac{2}{t^3}\mathbf{i} + \frac{3}{t^2}\mathbf{j} + 2t\mathbf{k}$$

$$P = (2, -2, 1) \Rightarrow t = 1$$

$$\mathbf{r}'(1) = -2\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$$

C: $\mathbf{r}'(1) \cdot \mathbf{v} = -2 \cdot (-1) + 3 \cdot (-2) + 2 \cdot 2 = 0$

D (2) The derivative of $f(x, y, z) = x^2y + z^2$ at the point $(1, 2, 0)$ in the direction of $\mathbf{v} = \langle 1, 2, 2 \rangle$ is

(A) 12.

(B) 6.

(C) 4.

(D) 2.

$$\nabla f = (2xy, x^2, 2z)$$

$$\nabla f(1, 2, 0) = (4, 1, 0)$$

$$D_{\mathbf{v}}f = \frac{\nabla f \cdot \mathbf{v}}{|\mathbf{v}|} = \frac{6}{\sqrt{1+4+4}} = 2$$

C (3) The series

$$\sum_{n=1}^{\infty} \left(1 - \frac{a}{n}\right)^{n^2}$$

converges

(A) for any real number a .

(B) for any $a \in (-1, 1)$.

(C) for any $a > 0$.

(D) for any $a < 0$.

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left|1 - \frac{a}{n}\right|^{n^2}} = \lim_{n \rightarrow \infty} \left(1 - \frac{a}{n}\right)^n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{\frac{n}{a}}\right)^{-\frac{n}{a}(-a)} = e^{-a} < 1$$

$$e^a > 1 \Rightarrow a > 0$$

1) (4) $\lim_{(x,y) \rightarrow (0,0)} (1 + \sin(xy))^{\frac{1}{x^2+y^2}} =$

(A) 0 .

(B) 1 .

(C) e .

(D) does not exist.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\ln(1 + \sin(xy))}{x^2 + y^2} = \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy)}{x^2 + y^2}$$

$y = kx$ $\lim_{(x,kx) \rightarrow (0,0)} \frac{kx^2 + \sin(kx^2)}{x^2 + k^2x^2} = \frac{k}{1+k^2}$ for different k , $\frac{k}{1+k^2}$ different.

1) (5) $\int_0^{\pi/4} \int_0^{2 \cos \theta} f(r \cos \theta, r \sin \theta) r dr d\theta =$

(A) $\int_0^1 \int_y^{\sqrt{2y-y^2}} f(x,y) dx dy.$

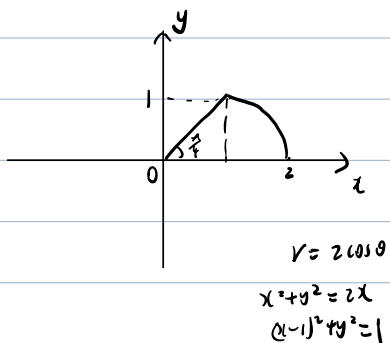
(B) $\int_0^1 \int_{1-\sqrt{1-y^2}}^y f(x,y) dx dy.$

(C) $\int_0^1 \int_{-x}^x f(x,y) dy dx + \int_1^2 \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} f(x,y) dy dx.$

(D) $\int_0^1 \int_0^x f(x,y) dy dx + \int_1^2 \int_0^{\sqrt{2x-x^2}} f(x,y) dy dx.$

$$(x-1)^2 + y^2 = 1$$

$$x-1 = \sqrt{1-y^2}$$



$$\int_0^1 \int_y^{1+\sqrt{1-y^2}} f(x,y) dx dy$$

$$= \int_0^1 \int_0^x f(x,y) dy dx + \int_1^2 \int_0^{\sqrt{2x-x^2}} f(x,y) dy dx$$

2. (20 pts) Fill in the blanks.

(1) The equation of plane tangent to the surface $xyz + e^{x+z} = 0$ at the point $(1, 1, -1)$ is

$$y - 2z - 3 = 0$$

$$F_x = yz + e^{x+z} \quad F_y = xz \quad F_z = xy + e^{x+z}$$

$$F_x|_{(1,1,-1)} = 0 \quad F_y|_{(1,1,-1)} = -1 \quad F_z|_{(1,1,-1)} = 2$$

$$0(x-1) - (y-1) + 2(z+1) = 0$$

(2) If the Maclaurin series of a function $f(x)$ is $\sum_{n=1}^{\infty} \frac{x^{2n+2}}{n(n+4)2^n}$, then $f^{(6)}(0) = 15$.

$$n=2 \quad \frac{f^{(6)}(0)}{6!} = \frac{1}{2(2+4)2^2} \quad f^{(6)}(0) = \frac{5!}{8} = 15$$

(3) The curvature κ for the curve $\mathbf{r}(t) = e^t \cos t \mathbf{i} + e^t \sin t \mathbf{j} + 2\mathbf{k}$ is $\frac{\sqrt{2}}{2} e^{-t}$.

$$\kappa = \left| \frac{dT}{ds} \right| = \left| \frac{dT/dt}{ds/dt} \right| = \frac{1}{|v|} \left| \frac{dT}{dt} \right|$$

$$\frac{dv}{dt} = e^t (\cos t - \sin t) \mathbf{i} + e^t (\sin t + \cos t) \mathbf{j}$$

$$|v| = \left| \frac{dv}{dt} \right| = \sqrt{2} e^t$$

$$T = \frac{dv}{dt} / \left| \frac{dv}{dt} \right| = \frac{\sqrt{2}}{2} (\cos t - \sin t) \mathbf{i} + \frac{\sqrt{2}}{2} (\sin t + \cos t) \mathbf{j}$$

$$\frac{dT}{dt} = \frac{\sqrt{2}}{2} (-\sin t - \cos t) \mathbf{i} + \frac{\sqrt{2}}{2} (\cos t - \sin t) \mathbf{j} \quad \left| \frac{dv}{dt} \right| = 1$$

$$\kappa = \left| \frac{dT}{dt} \right| \frac{1}{|v|} = \frac{1}{\sqrt{2} e^t} = \frac{\sqrt{2}}{2} e^{-t}$$

(4) If a function $y = y(x)$ is defined by the parametric equations

$$x = 2t + \ln^2 t, \quad y = (t + \ln t)^2, \quad t > 0,$$

$$\text{then } \left. \frac{d^2 y}{dx^2} \right|_{t=1} = \frac{1}{2}.$$

$$\frac{dy}{dt} = 2(t + \ln t) \left(1 + \frac{1}{t}\right)$$

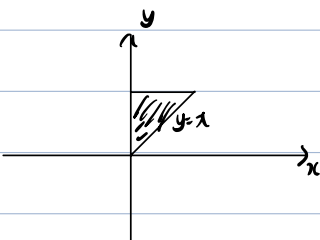
$$\frac{dx}{dt} = 2 + 2 \ln t \cdot \frac{1}{t}$$

$$\frac{dy}{dx} = \frac{(t + \ln t) \left(1 + \frac{1}{t}\right)}{1 + \ln t \cdot \frac{1}{t}} = \frac{(t + \ln t)(t + 1)}{t + \ln t} = t + 1$$

$$\left(\frac{dy}{dx} \right) / dt = 1$$

$$\left. \frac{d^2 y}{dx^2} \right|_{t=1} = \left(\frac{dy}{dx} \right) / dt \bigg/ \frac{dx}{dt} = \frac{1}{2(t + \ln t)} \bigg|_{t=1} = \frac{1}{2}$$

(5) $\int_0^1 \int_x^1 x e^{-y^3} dy dx = \frac{1}{6} - \frac{1}{6e}.$



$$\int_0^1 \int_x^1 x e^{-y^3} dy dx = \int_0^1 \int_0^y x e^{-y^3} dx dy$$

$$= \int_0^1 \frac{1}{2} x^2 e^{-y^3} \bigg|_0^y dy = \int_0^1 \frac{1}{2} y^2 e^{-y^3} dy = -\frac{1}{6} e^{-y^3} \bigg|_0^1 = -\frac{1}{6} (e^{-1} - 1) = \frac{1}{6} - \frac{1}{6e}$$

3. (10 pts) Find all the local extreme values of $f(x, y) = x^3 + 2y^2 - 3x - 12y$.

$$f_x = 3x^2 - 3 \quad f_{xx} = 6x$$

$$f_y = 4y - 12 \quad f_{yy} = 4$$

$$f_{xy} = 0$$

$$\begin{cases} f_x = 0 \\ f_y = 0 \end{cases} \Rightarrow \begin{cases} x = 1 \\ y = 3 \end{cases} \quad \text{or} \quad \begin{cases} x = -1 \\ y = 3 \end{cases}$$

$$\textcircled{1} \quad f_{xx}(1, 3) = 6 > 0$$

$$(f_{xx}f_{yy} - f_{xy}^2)(1, 3) = 6 \cdot 4 - 0 = 24 > 0$$

$$f(1, 3) = 1 + 2 \cdot 9 - 3 \cdot 1 - 12 \cdot 3 = -20$$

f has a local minimum at $(1, 3)$

$$\textcircled{2} \quad f_{xx}(-1, 3) = -6 < 0$$

$$(f_{xx}f_{yy} - f_{xy}^2)(-1, 3) = -6 \cdot 4 - 0 = -24 < 0$$

f has a saddle point at $(-1, 3)$

4. (10 pts)

(1) Find the interval of convergence of the series $\sum_{n=2}^{\infty} \frac{(-1)^n (x-1)^{2n+1}}{(n+2023) \ln n}$.

(2) For what values of x does the series converge absolutely, or conditionally?

$$(1) \quad \text{For } n \text{ big} \quad 1 \leq \sqrt[n]{\ln n} \leq \sqrt[n]{n} \quad \lim_{n \rightarrow \infty} \sqrt[n]{\ln n} = 1$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{(-1)^n (x-1)^{2n+1}}{(n+2023) \ln n} \right|} = \lim_{n \rightarrow \infty} \frac{|x-1|^{2+\frac{1}{n}}}{\sqrt[n]{n+2023} \sqrt[n]{\ln n}} = |x-1|^2$$

$$|x-1|^2 < 1 \Rightarrow x \in (0, 2)$$

$$\sum_{n=2}^{\infty} \frac{(-1)^n (x-1)^{2n+1}}{(n+2023) \ln n} \text{ converges absolutely}$$

$$x-1=1 \Rightarrow x=2$$

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{(n+2023) \ln n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{(n+2023) \ln n} = 0$$

$$\frac{1}{(n+2023) \ln n} > 0$$

$$f(x) = \frac{1}{(x+2023) \ln x}$$

$$f'(x) = -\frac{\ln x + 1 + \frac{2023}{x}}{(x+2023)^2 (\ln x)^2} < 0 \quad \text{for } x \geq 2$$

$\frac{1}{(n+2023) \ln n}$ is non-increasing for $n \geq 2$.

By alternating series test, $\sum_{n=2}^{\infty} \frac{(-1)^n}{(n+2023) \ln n}$ converges

$$x-1 = -1 \quad \sum_{n=2}^{\infty} \frac{(-1)^n (-1)^{2n+1}}{(n+2022) \ln n} = - \sum_{n=2}^{\infty} \frac{(-1)^n}{(n+2022) \ln n} \quad \text{converges}$$

$$\text{So } x \in [0, 2] \quad \sum_{n=2}^{\infty} \frac{(-1)^n x^{2n+1}}{(n+2022) \ln n} \quad \text{converges}$$

$$(2) \quad \sum_{n=2}^{\infty} \frac{1}{(n+2022) \ln n}$$

$$\frac{1}{(n+2022) \ln n} > \frac{1}{(n+2022) \ln(n+2022)}$$

$$f(x) = \frac{1}{x \ln x} \quad \text{is non-increasing for } x \geq 2.$$

By integral test for series.

$$\int_2^{\infty} \frac{1}{x \ln x} dx \quad \text{and} \quad \sum_{n=2}^{\infty} \frac{1}{x \ln x} \quad \text{both converge or diverge.}$$

$$\int_2^{\infty} \frac{1}{x \ln x} dx = \infty$$

$$\text{So } \sum_{n=2}^{\infty} \frac{1}{n \ln n} \quad \text{is diverges.}$$

$$\text{By comparison Test, } \sum_{n=2}^{\infty} \frac{1}{(n+2022) \ln n} \quad \text{diverges}$$

$$\text{So } x=0 \quad \text{or } 1 \quad \sum_{n=2}^{\infty} \frac{(-1)^n x^{2n+1}}{(n+2022) \ln n} \quad \text{converges conditionally.}$$

$$x \in (0, 1) \quad \sum_{n=2}^{\infty} \frac{(-1)^n x^{2n+1}}{(n+2022) \ln n} \quad \text{converges absolutely.}$$

5. (10 pts) Find the center of mass of the hemisphere $D: x^2 + y^2 + z^2 \leq 1, z \geq 0$ with the density function $\delta(x, y, z) = z$.

$$\text{Let } \begin{cases} x = \rho \sin \phi \cos \theta & 0 \leq \rho \leq 1 \\ y = \rho \sin \phi \sin \theta & 0 \leq \phi \leq \frac{\pi}{2} \\ z = \rho \cos \phi & 0 \leq \theta \leq 2\pi \end{cases}$$

$$M = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^1 z \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^1 \rho^3 \sin \phi \cos \phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} \frac{1}{2} \sin 2\phi \, d\phi \int_0^1 \rho^4 \, d\rho$$

$$= 2\pi \left[-\frac{1}{4} \cos 2\phi \right]_0^{\frac{\pi}{2}} \left[\frac{1}{5} \rho^5 \right]_0^1$$

$$= \frac{\pi}{5}$$

$$M_{yz} = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^1 xz \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^1 \rho^4 \sin^2\phi \cos\phi \cos\theta \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \cos\theta \, d\theta \int_0^{\frac{\pi}{2}} \sin^2\phi \cos\phi \, d\phi \int_0^1 \rho^4 \, d\rho$$

$$= \sin\theta \Big|_0^{2\pi} \int_0^{\frac{\pi}{2}} \sin^2\phi \cos\phi \, d\phi \int_0^1 \rho^4 \, d\rho$$

$$= 0$$

$$M_{zx} = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^1 yz \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^1 \rho^4 \sin^2\phi \cos\phi \sin\theta \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \sin\theta \, d\theta \int_0^{\frac{\pi}{2}} \sin^2\phi \cos\phi \, d\phi \int_0^1 \rho^4 \, d\rho$$

$$= -\cos\theta \Big|_0^{2\pi} \int_0^{\frac{\pi}{2}} \sin^2\phi \cos\phi \, d\phi \int_0^1 \rho^4 \, d\rho$$

$$= 0$$

$$M_{xy} = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^1 z^2 \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_0^1 \rho^4 \cos^2\phi \sin\phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} d\theta \left(\int_0^{\frac{\pi}{2}} \cos^2\phi \, d(\cos\phi) \right) \int_0^1 \rho^4 \, d\rho$$

$$= 2\pi \left(-\frac{1}{3} \cos^3\phi \right) \Big|_0^{\frac{\pi}{2}} \frac{1}{5} \rho^5 \Big|_0^1$$

$$= 2\pi \cdot \left(-\frac{1}{3}\right) \cdot \frac{1}{5} = -\frac{2\pi}{15}$$

$$\bar{x} = \frac{M_{yz}}{m} = 0 \quad \bar{y} = \frac{M_{zx}}{m} = 0 \quad \bar{z} = \frac{M_{xy}}{m} = -\frac{8}{15}$$

The center of mass is $(0, 0, -\frac{8}{15})$

6. (10 pts) Given the conservative vector field

$$\mathbf{F} = (y \sin z + 2)\mathbf{i} + (x \sin z)\mathbf{j} + (xy \cos z)\mathbf{k},$$

find its potential function and evaluate the integral

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

along any smooth curve C joining point $A(1, -2, \pi/3)$ to $B(2, 1, \pi/4)$.

Let f be function s.t. $df = \mathbf{F}$

($\mathbf{F}$ is conservative $\Rightarrow f$ always exists.)

$$\frac{\partial f}{\partial x} = y \sin z + 2$$

$$f = xy \sin z + 2x + f_1(y, z)$$

$$\frac{\partial f}{\partial y} = x \sin z + \frac{\partial f_1}{\partial y} = x \sin z$$

$$\frac{\partial f_1}{\partial y} = 0 \Rightarrow f_1 = f_2(z)$$

$$\Rightarrow f = xy \sin z + 2x + f_2(z)$$

$$\frac{\partial f}{\partial z} = xy \cos z + \frac{\partial f_2}{\partial z} = xy \cos z \Rightarrow \frac{\partial f_2}{\partial z} = 0 \quad f_2 = C$$

$$\text{So } f = xy \sin z + 2x + C$$

So for any smooth curve from $(1, -2, \frac{\pi}{3})$ to $(2, 1, \frac{\pi}{4})$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C df = f(2, 1, \frac{\pi}{4}) - f(1, -2, \frac{\pi}{3})$$

$$= 2 \sin \frac{\pi}{4} + 4 - (-2 \sin \frac{\pi}{3} + 2)$$

$$= \sqrt{2} + \sqrt{3} + 2$$

7. (10 pts) The vector-field

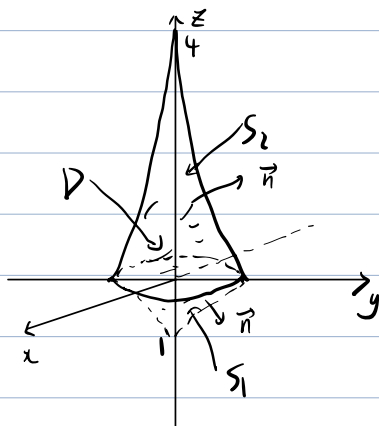
$$\mathbf{F}(x, y, z) = (z + xy^2)\mathbf{i} + (2y + y^3)\mathbf{j} - 4zy^2\mathbf{k}$$

is defined in the domain V bounded by the surfaces S_1 and S_2 ,

$$S_1 := \{(x, y, z) : z = (x^2 + y^2)^2 - 1\},$$

$$S_2 := \{(x, y, z) : z = 4 - 4(x^2 + y^2)\}.$$

Apply the **Divergence Theorem** to compute the outward flux of $\mathbf{F}$ across the boundary of V .



$$\begin{aligned}\nabla \cdot \mathbf{F} &= \frac{\partial (z + xy^2)}{\partial x} + \frac{\partial (2y + y^3)}{\partial y} + \frac{\partial (-4zy^2)}{\partial z} \\ &= y^2 + 2 + 3y^2 - 4y^2 = 2\end{aligned}$$

By Divergence Theorem,

$$\begin{aligned}\iint_{S_1 \cup S_2} \mathbf{F} \cdot \vec{n} \, d\sigma &= \iiint_V \nabla \cdot \mathbf{F} \, dV \\ &= 2 \iiint_V dV\end{aligned}$$

$$\text{Let } \begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases} \quad \begin{cases} r^4 - 1 \leq z \leq 4 - 4r^2 \\ 0 \leq r \leq 1 \\ 0 \leq \theta \leq 2\pi \end{cases}$$

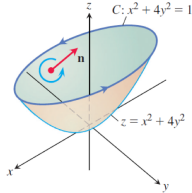
$$\begin{aligned}&= 2 \int_0^{2\pi} \int_0^1 \int_{r^4-1}^{4-4r^2} r \, dz \, dr \, d\theta \\ &= 2 \int_0^{2\pi} d\theta \int_0^1 r(4 - 4r^2 - r^4 + 1) \, dr \\ &= 2 \cdot 2\pi \int_0^1 (-r^5 - 4r^3 + 5r) \, dr \\ &= 4\pi \left(-\frac{r^6}{6} - r^4 + \frac{5}{2}r^2 \right) \Big|_0^1 \\ &= 4\pi \left(-\frac{1}{6} - 1 + \frac{5}{2} \right) = \frac{16\pi}{3}\end{aligned}$$

8. (10 pts) Let the surface S be the elliptical paraboloid $z = x^2 + 4y^2$ lying beneath the plane $z = 1$ (Figure is given below). We define the orientation of S by taking the inner normal vector $\mathbf{n}$ to the surface, which is the normal having a positive k -component. Compute

$$\iint_S \nabla \times \mathbf{F} \cdot \mathbf{n} \, d\sigma,$$

here

$$\mathbf{F} = (y + x^2 \ln(x^4 + 1))\mathbf{i} + (e^{\sin y} - xz)\mathbf{j} + (xz^2 + \cos(z^2 + 1))\mathbf{k}.$$



$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y + x^2 \ln(x^4 + 1) & e^{\sin y} - xz & xz^2 + \cos(z^2 + 1) \end{vmatrix}$$

$$= -x\mathbf{i} - z^2\mathbf{j} + (-z-1)\mathbf{k}$$

$$= -x\mathbf{i} - z^2\mathbf{j} - (z+1)\mathbf{k}$$

Let $S_1 = \{(x, y) \mid x^2 + 4y^2 \leq 1\}$ the normal vector of S : $\vec{n}_1 = (0, 0, 1)$

$$\nabla \times \mathbf{F} \cdot \vec{n}_1 = -(z+1)$$

By Stokes' Theorem,

$$\iint_S \nabla \times \mathbf{F} \cdot \vec{n} \, d\sigma = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_{S_1} \nabla \times \mathbf{F} \cdot \vec{n}_1 \, d\sigma$$

$$= \iint_{S_1} -(z+1) \, d\sigma$$

$$= - \iint_{S_1} d\sigma = -\pi \cdot 1 \cdot \left(\frac{1}{2}\right) = -\frac{\pi}{2}$$